

```

riya59@Priyanka:~/ledger-api-assignment$ docker run -d \
--name ledger-api-test \
-p 8080:8080 \
ledger-api:starter
770fbf1dfb08796cfa203b74dbbc104ef931c851ecea93f7c0e3b610eed971
riya59@Priyanka:~/ledger-api-assignment$ docker ps
CONTAINER ID   IMAGE                                COMMAND                  CREATED        STATUS        PORTS
770fbf1dfb0    ledger-api:starter                 "python app.py"         22 seconds ago Up 21 seconds 0.0.0.0:8080->8080/tcp, [::]:8080->80
5a4d6601d86    kindest/node:v1.34.0              "/usr/local/bin/entr..." 4 minutes ago  Up 4 minutes 127.0.0.1:42553->6443/tcp
3b9e13b3909    kindest/node:v1.34.0              "/usr/local/bin/entr..." 4 minutes ago  Up 4 minutes 127.0.0.1:42553->6443/tcp
riya59@Priyanka:~/ledger-api-assignment$ curl http://localhost:8080/health

{"status": "ok"}

riya59@Priyanka:~/ledger-api-assignment$ curl -X POST http://localhost:8080/tokenize \
-H "Content-Type: application/json" \
-d '{"pan": "4242424242424242"}'

{"last4": "4242",
"token": "tok_477bba133c182267fe5f0869"}

riya59@Priyanka:~/ledger-api-assignment$ docker exec ledger-api-test id
id=0(root) gid=0(root) groups=0(root)
riya59@Priyanka:~/ledger-api-assignment$ docker inspect ledger-api-test \
--format='ReadOnlyRootfs={{.HostConfig.ReadOnlyRootfs}}'
ReadOnlyRootfs=false
riya59@Priyanka:~/ledger-api-assignment$

```

ledger-api:starter

b2ca1653c051

```
priya59@Priyanka:~/ledger-api-assignment$ docker run -d \
--name ledger-api-test \
-p 8080:8080 \
ledger-api:starter
```

7770fbf1dfb08796cfa203b74dbbc104ef931c851eceaa93f7c0e3b610eed971

```
priya59@Priyanka:~/ledger-api-assignment$ docker ps
```

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
7770fbf1dfb0	ledger-api:starter	"python app.py"	22 seconds ago	Up 21 seconds	0.0.0.0:8080->8080/tcp
15a4d6601d86	kindest/node:v1.34.0	"/usr/local/bin/entr..."	4 minutes ago	Up 4 minutes	
23b9e13b3909	kindest/node:v1.34.0	"/usr/local/bin/entr..."	4 minutes ago	Up 4 minutes	127.0.0.1:2345->2345/tcp

```
priya59@Priyanka:~/ledger-api-assignment$ curl http://localhost:8080/health
```

```
{
  "status": "ok"
}
```

```
priya59@Priyanka:~/ledger-api-assignment$ curl -X POST http://localhost:8080/tokenize \
```

```

pod/ledger-api-7cd8477dfc-wj5qq 1/1 Running 0 6m1s
pod/reporting-684f6b9c8b-strnt 1/1 Running 0 6m

NAME          TYPE          CLUSTER-IP    EXTERNAL-IP    PORT(S)    AGE
service/ledger-api ClusterIP      10.96.4.33    <none>         8080/TCP    6m

NAME          READY    UP-TO-DATE    AVAILABLE    AGE
deployment.apps/ledger-api 3/3      3              3             6m1s
deployment.apps/reporting  1/1      1              1             6m

NAME          DESIRED    CURRENT    READY    AGE
replicaset.apps/ledger-api-7cd8477dfc 3          3          3        6m1s
replicaset.apps/reporting-684f6b9c8b 1          1          1         6m
priya59@Priyanka:~/ledger-api-assignment$ kubectl exec -n payments curl-test -- curl -s http://ledger-api:8080/health
{
  "status": "ok"
}
priya59@Priyanka:~/ledger-api-assignment$ kubectl exec -n payments <ledger-pod> -- id
-bash: ledger-pod: No such file or directory
priya59@Priyanka:~/ledger-api-assignment$ kubectl get pod <ledger-pod> -n payments -o jsonpath='{.spec.serviceAccountName}'
-bash: ledger-pod: No such file or directory
priya59@Priyanka:~/ledger-api-assignment$ kubectl get pods -n payments
NAME          READY    STATUS    RESTARTS    AGE
curl-test     1/1      Running   0            5m50s
ledger-api-7cd8477dfc-blvcc 1/1      Running   0            7m29s
ledger-api-7cd8477dfc-ctfpp 1/1      Running   0            7m29s
ledger-api-7cd8477dfc-wj5qq 1/1      Running   0            7m29s
reporting-684f6b9c8b-strnt 1/1      Running   0            7m28s
priya59@Priyanka:~/ledger-api-assignment$ kubectl exec -n payments ledger-api-7cd8477dfc-blvcc -- id
uid=0(root) gid=0(root) groups=0(root)
priya59@Priyanka:~/ledger-api-assignment$ kubectl get pod ledger-api-7cd8477dfc-blvcc -n payments -o jsonpath='{.spec.serviceAccountName}'
default
priya59@Priyanka:~/ledger-api-assignment$ |

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```

priya59@Priyanka:~/ledger-api-assignment$ docker run -d \
  --name ledger-api-hardened-test \
  -p 8081:8080 \
  ledger-api:hardened
29264d0991a474b933d14a6aa7d365fa1c2d17a06751535ef1ab4443aeb21f9
priya59@Priyanka:~/ledger-api-assignment$ docker ps --filter name=ledger-api-hardened-test
NAME                  CONTAINER ID   IMAGE                  COMMAND                  CREATED        STATUS        PORTS
ledger-api-hardened-test 29264d0991a    ledger-api:hardened   "python app.py"         9 seconds ago Up 9 seconds   0.0.0.0:8081->8080/tcp, [::]:8081->8080/tcp
* Serving Flask app 'app'
* Debug mode: off
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:8080
* Running on http://172.17.0.2:8080
Press CTRL+C to quit
priya59@Priyanka:~/ledger-api-assignment$ curl http://localhost:8081/health
{"status":"ok"}
priya59@Priyanka:~/ledger-api-assignment$ docker exec ledger-api-hardened-test id
uid=10001(appuser) gid=10001 groups=10001
priya59@Priyanka:~/ledger-api-assignment$

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priya59@Priyanka:~/ledger-api-assignment$ kubectl apply -f task-1/hardened/deployment.yaml
deployment.apps/ledger-api configured
priya59@Priyanka:~/ledger-api-assignment$ kubectl rollout status deployment/ledger-api -n payments
Waiting for deployment "ledger-api" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "ledger-api" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "ledger-api" rollout to finish: 1 out of 3 new replicas have been updated...
Waiting for deployment "ledger-api" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "ledger-api" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "ledger-api" rollout to finish: 2 out of 3 new replicas have been updated...
Waiting for deployment "ledger-api" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "ledger-api" rollout to finish: 1 old replicas are pending termination...
deployment "ledger-api" successfully rolled out
priya59@Priyanka:~/ledger-api-assignment$ kubectl get pods -n payments
NAME                                READY   STATUS    RESTARTS   AGE
curl-test                          1/1     Running   0           23m
ledger-api-7cd8477dfc-blvcc        1/1     Terminating   0           25m
ledger-api-7cd8477dfc-wj5qq        1/1     Terminating   0           25m
ledger-api-8599886546-g64rr        1/1     Running        0           49s
ledger-api-8599886546-n49j8        1/1     Running        0           23s
ledger-api-8599886546-whddd        1/1     Running        0           35s
reporting-684f6b9c8b-strnt         1/1     Running        0           25m
priya59@Priyanka:~/ledger-api-assignment$ LEDGER_POD=$(kubectl get pod -n payments -l app=ledger-api -o jsonpath='{.items[0].metadata.name}')

kubectl exec -n payments "$LEDGER_POD" -- id
kubectl get pod "$LEDGER_POD" -n payments -o jsonpath='{.spec.serviceAccountName}{"\n"}'
kubectl get pod "$LEDGER_POD" -n payments -o jsonpath='{.spec.securityContext.seccompProfile.type}{"\n"}'
error: Internal error occurred: Internal error occurred: error executing command in container: failed to exec in container: failed to start exec "6772e8fa8d7df4b0aa792a07da0b6fbe6f2a4bd608771f3cd2191cceaee82464": container "7598da9ec52b035289f5a3bbbe8a4780a192813bb10758a5b3d0d089d49413cc" init process is not running: failed precondition
default

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```

priya59@Priyanka:~/ledger-api-assignment$ kubectl get pods -n payments
NAME                                READY   STATUS    RESTARTS   AGE
curl-test                          1/1     Running   0           25m
ledger-api-8599886546-g64rr        1/1     Running   0           2m44s
ledger-api-8599886546-n49j8        1/1     Running   0           2m18s
ledger-api-8599886546-whddd         1/1     Running   0           2m30s
reporting-684f6b9c8b-strnt         1/1     Running   0           26m
priya59@Priyanka:~/ledger-api-assignment$ LEDGER_POD=$(kubectl get pods -n payments \
-l app=ledger-api \
-o json | jq -r \
'.items[] | select(.metadata.deletionTimestamp == null and .status.phase == "Running") | .metadata.name' \
| head -n 1)
echo "$LEDGER_POD"
ledger-api-8599886546-g64rr
priya59@Priyanka:~/ledger-api-assignment$ kubectl exec -n payments "$LEDGER_POD" -- id
uid=10001(appuser) gid=10001 groups=10001
priya59@Priyanka:~/ledger-api-assignment$ kubectl get pod "$LEDGER_POD" \
-n payments \
-o jsonpath='{.spec.serviceAccountName}'"
ledger-api-sa
priya59@Priyanka:~/ledger-api-assignment$ kubectl get pod "$LEDGER_POD" \
-n payments \
-o jsonpath='{.spec.securityContext.seccompProfile.type}'"
RuntimeDefault
priya59@Priyanka:~/ledger-api-assignment$ kubectl get pod "$LEDGER_POD" -n payments \
-o jsonpath='runAsNonRoot={.spec.securityContext.runAsNonRoot}'"allowPrivilegeEscalation={.spec.containers[0].securityContext.a
llowPrivilegeEscalation}'"readOnlyRootFilesystem={.spec.containers[0].securityContext.readOnlyRootFilesystem}'"
runAsNonRoot=true
allowPrivilegeEscalation=false
readOnlyRootFilesystem=true
priya59@Priyanka:~/ledger-api-assignment$ |

```

```
priya59@Priyanka:~/ledger-api-assignment$ cat > task-1/hardened/networkpolicy.yaml
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: ledger-api-network-policy
  namespace: payments
spec:
  podSelector:
    matchLabels:
      app: ledger-api

  policyTypes:
    - Ingress
    - Egress

  ingress:
    - from:
        - podSelector:
            matchLabels:
              app: reporting
      ports:
        - protocol: TCP
          port: 8080

  egress:
    - {}
EOF
priya59@Priyanka:~/ledger-api-assignment$ kubectl apply -f task-1/hardened/network
networkpolicy.networking.k8s.io/ledger-api-network-policy created
priya59@Priyanka:~/ledger-api-assignment$ kubectl get networkpolicy -n payments
NAME                                POD-SELECTOR    AGE
```



```
priya59@Priyanka:~/ledger-api-assignment$ cat > task-1/hardened/networkpolicy.yaml <<'EOF'
```

```
apiVersion: networking.k8s.io/v1
kind: NetworkPolicy
metadata:
  name: ledger-api-network-policy
  namespace: payments
```

```
spec:
  podSelector:
    matchLabels:
      app: ledger-api
```

```
  policyTypes:
    - Ingress
    - Egress
```

```
  ingress:
    - from:
        - podSelector:
            matchLabels:
              app: reporting
      ports:
        - protocol: TCP
          port: 8080
```

```
  egress:
    - {}
```

```
EOF
```

```
priya59@Priyanka:~/ledger-api-assignment$ kubectl apply -f task-1/hardened/networkpolicy.yaml
```

```
networkpolicy.networking.k8s.io/ledger-api-network-policy created
```

```
priya59@Priyanka:~/ledger-api-assignment$ kubectl get networkpolicy -n payments
```

```
NAME                                POD-SELECTOR    AGE
ledger-api-network-policy    app=ledger-api  8s
```

```
priya59@Priyanka:~/ledger-api-assignment$ |
```



```
priya59@Priyanka:~/ledger-api-assignment$ kubectl apply -f task-1/hardened/networkpolicy.yaml
networkpolicy.networking.k8s.io/ledger-api-network-policy created
priya59@Priyanka:~/ledger-api-assignment$ kubectl get networkpolicy -n payments
NAME                                POD-SELECTOR    AGE
ledger-api-network-policy    app=ledger-api  8s
priya59@Priyanka:~/ledger-api-assignment$ kubectl describe networkpolicy ledger-api-network-policy -n payments
Name:      ledger-api-network-policy
Namespace: payments
Created on: 2026-07-22 11:04:15 +0000 UTC
Labels:    <none>
Annotations: <none>
Spec:
  PodSelector:    app=ledger-api
  Allowing ingress traffic:
    To Port: 8080/TCP
    From:
      PodSelector: app=reporting
  Allowing egress traffic:
    To Port: <any> (traffic allowed to all ports)
    To: <any> (traffic not restricted by destination)
  Policy Types: Ingress, Egress
priya59@Priyanka:~/ledger-api-assignment$ |
```

```

priya59@Priyanka:~/ledger-api-assignment$ curl -H "Host: ledger.local" http://localhost:8082/health
{"status":"ok"}
priya59@Priyanka:~/ledger-api-assignment$ curl -X POST \
-H "Host: ledger.local" \
-H "Content-Type: application/json" \
http://localhost:8082/tokenize \
-d '{"pan":"4242424242424242"}'
{"last4":"4242","token":"tok_477bba133c182267fe5f0869"}
priya59@Priyanka:~/ledger-api-assignment$ kubectl get ingress -n payments \
> task-1/evidence/07-ingress.txt

kubectl describe ingress ledger-api-ingress -n payments \
> task-1/evidence/08-ingress-description.txt

curl -H "Host: ledger.local" http://localhost:8082/health \
| tee task-1/evidence/09-ingress-health.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %         0      Dload  Upload   Total   Spent    Left   Speed
100    16 100    16    0      0   1493      0      0      0
{"status":"ok"}
priya59@Priyanka:~/ledger-api-assignment$

```

```

priya59@Priyanka:~/ledger-api-assignment$ kubectl port-forward \
-n ingress-nginx \
service/ingress-nginx-controller \
8082:80 \
> /tmp/ledger-ingress.log 2>&1 &
[1] 13118
priya59@Priyanka:~/ledger-api-assignment$ curl -H "Host: ledger.local" http://localhost:8082/health
{"status":"ok"}
priya59@Priyanka:~/ledger-api-assignment$ curl -X POST \
-H "Host: ledger.local" \
-H "Content-Type: application/json" \
http://localhost:8082/tokenize \
-d '{"pan":"4242424242424242"}'
{"last4":"4242","token":"tok_477bba133c182267fe5f0869"}
priya59@Priyanka:~/ledger-api-assignment$ kubectl get ingress -n payments \
> task-1/evidence/07-ingress.txt

kubectl describe ingress ledger-api-ingress -n payments \
> task-1/evidence/08-ingress-description.txt

curl -H "Host: ledger.local" http://localhost:8082/health \
| tee task-1/evidence/09-ingress-health.txt
% Total    % Received % Xferd  Average Speed   Time    Time     Time  Current
           %             %             Dload  Upload  Total   Spent    Left   Speed
100    16 100    16    0      0   1493      0      0     0
{"status":"ok"}
priya59@Priyanka:~/ledger-api-assignment$ kill "$INGRESS_PID"
-bash: kill: ``: not a pid or valid job spec
priya59@Priyanka:~/ledger-api-assignment$ curl -H "Host: ledger.local" http://localhost:8082/health
{"status":"ok"}
priya59@Priyanka:~/ledger-api-assignment$ |

```

```
priya59@Priyanka:~/ledger-api-assignment$ mkdir -p task-1/kyverno
nano task-1/kyverno/require-nonroot.yaml
priya59@Priyanka:~/ledger-api-assignment$ kubectl apply -f task-1/kyverno/require-nonroot.yaml
clusterpolicy.kyverno.io/require-run-as-non-root created
priya59@Priyanka:~/ledger-api-assignment$ kubectl get clusterpolicy
NAME                                ADMISSION    BACKGROUND    READY    AGE    MESSAGE
require-run-as-non-root            true         true          True     7s     Ready
priya59@Priyanka:~/ledger-api-assignment$ kubectl apply -f task-1/insecure/deployment.yaml
Error from server: error when applying patch:
{"metadata":{"annotations":{"kubernetes.io/last-applied-configuration":"{\"apiVersion\":\"ap
\"metadata\":{\"annotations\":{},\"labels\":{\"app\":\"ledger-api\",\"name\":\"ledger-api\", \"namesp
eplicas\":3,\"selector\":{\"matchLabels\":{\"app\":\"ledger-api\"}},\"template\":{\"metadata\":{\"la
\\spec\":{\"containers\":[{\\env\":[{\\name\":\"STRIPE_API_KEY\", \\value\":\"sk_live_9f3a2b7c1e4d8RB
\\\", \\value\":\"P@ssw0rd123\"}], \\image\":\"ledger-api:starter\", \\imagePullPolicy\":\"IfNotPresent\\
\": [{\\containerPort\":8080}]}]}]}\\n\", \"labels\":{\"app\":\"ledger-api\"}, \"spec\":{\"template\":{\"spec\":{\"$
me\":\"ledger-api\"}, \"automountServiceAccountToken\":null, \"containers\": [{\\setElementOrder/env\": [{\\name
ASSWORD\"}], \\setElementOrder/ports\": [{\\containerPort\":8080}], \"env\": [{\\name\":\"STRIPE_API_KEY\", \"value
\"valueFrom\":null}, {\\name\":\"DB_PASSWORD\", \"value\":\"P@ssw0rd123\", \"valueFrom\":null}], \"image\":\"ledger-api
ame\":\"ledger-api\", \"ports\": [{\\containerPort\":8080, \"name\":null, \"protocol\":null}], \"readinessProbe\":null
t\":null, \"volumeMounts\":null}], \"securityContext\":null, \"serviceAccountName\":null, \"volumes\":null}}}]"}
to:
Resource: "apps/v1, Resource=deployments", GroupVersionKind: "apps/v1, Kind=Deployment"
Name: "ledger-api", Namespace: "payments"
for: "task-1/insecure/deployment.yaml": error when patching "task-1/insecure/deployment.yaml": admission
c-fail" denied the request:

resource Deployment/payments/ledger-api was blocked due to the following policies

require-run-as-non-root:
autogen-require-run-as-non-root: 'validation error: Containers must run as non-root.
rule autogen-require-run-as-non-root failed at path /spec/template/spec/securityContext/runAsNon
```

```

priya59@Priyanka:~/ledger-api-assignment$ mkdir -p task-1/kyverno
nano task-1/kyverno/require-nonroot.yaml
priya59@Priyanka:~/ledger-api-assignment$ kubectl apply -f task-1/kyverno/require-nonroot.yaml
clusterpolicy.kyverno.io/require-run-as-non-root created
priya59@Priyanka:~/ledger-api-assignment$ kubectl get clusterpolicy
NAME                ADMISSION  BACKGROUND  READY  AGE  MESSAGE
require-run-as-non-root  true        true        True   7s   Ready
priya59@Priyanka:~/ledger-api-assignment$ kubectl apply -f task-1/insecure/deployment.yaml
Error from server: error when applying patch:
{"metadata":{"annotations":{"kubernetes.io/last-applied-configuration":{"{"apiVersion":"apps/v1","kind":"Deployment","\
"metadata":{"annotations":{"},"labels":{"app":"ledger-api"},"name":"ledger-api","namespace":"payments"},"spec":{"r
eplicas":3,"selector":{"matchLabels":{"app":"ledger-api"},"template":{"metadata":{"labels":{"app":"ledger-api"},
"spec":{"containers":[{"env":[{"name":"STRIPE_API_KEY","value":"sk_live_9f3a2b7c1e4d8REDACTED"},"name":"DB_PASSWORD
","value":"P@ssw0rd123"}],"image":"ledger-api:starter","imagePullPolicy":"IfNotPresent"},"name":"ledger-api","ports
":[{"containerPort":8080}]}}}}\n"},"labels":{"app":"ledger-api"},"spec":{"template":{"spec":{"$setElementOrder/containers":[{"na
me":"ledger-api"},"automountServiceAccountToken":null,"containers":[{"$setElementOrder/env":[{"name":"STRIPE_API_KEY"},"name":"DB_P
ASSWORD"}],"$setElementOrder/ports":[{"containerPort":8080},"env":[{"name":"STRIPE_API_KEY","value":"sk_live_9f3a2b7c1e4d8REDACTED",
"valueFrom":null},"name":"DB_PASSWORD","value":"P@ssw0rd123","valueFrom":null},"image":"ledger-api:starter","livenessProbe":null,"n
ame":"ledger-api","ports":[{"containerPort":8080,"name":null,"protocol":null},"readinessProbe":null,"resources":null,"securityContex
t":null,"volumeMounts":null},"securityContext":null,"serviceName":null,"volumes":null}}}}
to:
Resource: "apps/v1, Resource=deployments", GroupVersionKind: "apps/v1, Kind=Deployment"
Name: "ledger-api", Namespace: "payments"
for: "task-1/insecure/deployment.yaml": error when patching "task-1/insecure/deployment.yaml": admission webhook "validate.kyverno.sv
c-fail" denied the request:

resource Deployment/payments/ledger-api was blocked due to the following policies

require-run-as-non-root:
  autogen-require-run-as-non-root: 'validation error: Containers must run as non-root.
    rule autogen-require-run-as-non-root failed at path /spec/template/spec/securityContext/runAsNonRoot/'
priya59@Priyanka:~/ledger-api-assignment$ |

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```
push-to-registry: true
priya59@Priyanka:~/ledger-api-assignment$ cat task-2/Dockerfile
FROM python:3.12-slim

ENV PYTHONDONTWRITEBYTECODE=1
ENV PYTHONUNBUFFERED=1

WORKDIR /app

RUN groupadd --system appgroup && \
    useradd --system --gid appgroup --uid 10001 appuser

COPY app/requirements.txt .

RUN pip install --no-cache-dir -r requirements.txt

COPY app/app.py .

RUN chown -R appuser:appgroup /app

USER 10001:10001

EXPOSE 8080

CMD ["gunicorn", "--bind", "0.0.0.0:8080", "--workers", "2", "app:app"]
priya59@Priyanka:~/ledger-api-assignment$ |
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```
ID ["gunicorn", "--bind", "0.0.0.0:8080", "--workers", "2", "app:app"]
Priya59@Priyanka:~/ledger-api-assignment$ nano .github/workflows/build.yml
Priya59@Priyanka:~/ledger-api-assignment$ grep -n "task-2\|context:\|scan-ref:" .github/workflows/build.yml
6:         task-2/app/
7:         scan-ref: task-2
11:        context: task-2
12:        file: task-2/Dockerfile
14:        context: .
Priya59@Priyanka:~/ledger-api-assignment$ nano .github/workflows/build.yml
Priya59@Priyanka:~/ledger-api-assignment$ grep -n "task-2\|context:\|scan-ref:" .github/workflows/build.yml
6:         task-2/app/
7:         scan-ref: task-2
11:        context: task-2
12:        file: task-2/Dockerfile
14:        context: task-2
15:        file: task-2/Dockerfile
Priya59@Priyanka:~/ledger-api-assignment$ |
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        subject-digest: ${ steps.build.outputs.digest }
        push-to-registry: true
priya59@Priyanka:~/ledger-api-assignment$ git status
On branch main
Your branch is up to date with 'origin/main'.

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   app/app.py
        modified:   task-1/hardened/Dockerfile

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        task-2/evidence/

no changes added to commit (use "git add" and/or "git commit -a")
priya59@Priyanka:~/ledger-api-assignment$ |
```

```
5278b77..1a9cd13  main -> main
priya59@Priyanka:~/ledger-api-assignment$ git log -1 --stat | tee task-2/evidence/final-commit.txt
commit 1a9cd159926c8e25c374b23decc0e3cc84f0350e
Author: priya-cmd624 <priyankabandela59@gmail.com>
Date:   Wed Jul 22 16:53:41 2026 +0000

    Document validated SSRF protection for Semgrep

task-2/app/app.py | 3 +++
1 file changed, 3 insertions(+)
priya59@Priyanka:~/ledger-api-assignment$ cat .github/workflows/build.yml | tee task-2/evidence/github-actions-work
t
name: Secure CI/CD Pipeline

on:
```

priya-cmd624 / -ledger-api-assignment

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Secure CI/CD Pipeline

Document validated SSRF protection for Semgrep #5

Re-run all jobs

Summary

All jobs

✔ Gitleaks Secret Scan

✔ Semgrep SAST

✔ Trivy Dependency Scan

✔ Build, Scan, Sign and Attest

Run details

Usage

Workflow file

Triggered via push 7 minutes ago

Status

Total duration

Artifacts

priya-cmd624 pushed

1a9cd15

main

Success

1m 51s

3

build.yml

on: push

✔ Gitleaks Secret Scan

6s

✔ Semgrep SAST

19s

✔ Trivy Dependency Scan

14s

✔ Build, Scan, Sign and Att...

1m 24s

```

payments      reporting-684f6b9c8b-strnt      1/1      Running      1 (2m01s ago)      6h42m
priya59@Priyanka:~/ledger-api-assignment$ mkdir -p task-3/evidence

kubectl get nodes | tee task-3/evidence/cluster-nodes-before-istio.txt

kubectl get pods -A | tee task-3/evidence/cluster-pods-before-istio.txt

kubectl config current-context | tee task-3/evidence/kube-context.txt
NAME                                STATUS    ROLES    AGE      VERSION
dodo-security-control-plane        Ready    control-plane    6h53m    v1.34.0
dodo-security-worker               Ready    <none>          6h52m    v1.34.0

NAMESPACE      NAME                                                    READY    STATUS    RESTARTS      AGE
ingress-nginx   ingress-nginx-controller-f5784567-2mrzc                1/1      Running    1 (3m45s ago)  6h8m
kube-system     coredns-66bc5c9577-kd2hm                              1/1      Running    1 (3m45s ago)  6h52m
kube-system     coredns-66bc5c9577-p7pnz                              1/1      Running    1 (3m45s ago)  6h52m
kube-system     etcd-dodo-security-control-plane                      1/1      Running    1 (3m45s ago)  6h53m
kube-system     kindnet-4sr4p                                           1/1      Running    1 (3m45s ago)  6h52m
kube-system     kindnet-9ksbc                                           1/1      Running    1 (3m45s ago)  6h52m
kube-system     kube-apiserver-dodo-security-control-plane             1/1      Running    1 (3m45s ago)  6h53m
kube-system     kube-controller-manager-dodo-security-control-plane    1/1      Running    1 (3m45s ago)  6h53m
kube-system     kube-proxy-2mmp9                                        1/1      Running    1 (3m45s ago)  6h52m
kube-system     kube-proxy-qk656                                        1/1      Running    1 (3m45s ago)  6h52m
kube-system     kube-scheduler-dodo-security-control-plane             1/1      Running    1 (3m45s ago)  6h53m
kyverno         kyverno-admission-controller-6c596f64db-qtxsk         1/1      Running    1 (3m45s ago)  6h3m
kyverno         kyverno-background-controller-65885d77db-h2s2k        1/1      Running    1 (3m45s ago)  6h3m
kyverno         kyverno-cleanup-controller-568ff47f67-2w59k           1/1      Running    1 (3m45s ago)  6h3m
kyverno         kyverno-reports-controller-666865d4d-tsvd4            1/1      Running    1 (3m45s ago)  6h3m
local-path-storage local-path-provisioner-7b8c8ddb6-59g54                1/1      Running    2 (3m8s ago)   6h52m
payments        curl-test                                               0/1      Completed    0              6h41m
payments        ledger-api-8599886546-g64rr                           1/1      Running    1 (3m45s ago)  6h18m
payments        ledger-api-8599886546-n49j8                           1/1      Running    1 (3m45s ago)  6h18m
payments        ledger-api-8599886546-whddd                            1/1      Running    1 (3m45s ago)  6h18m
payments        reporting-684f6b9c8b-strnt                            1/1      Running    1 (3m45s ago)  6h43m

```

Events: None

priya59@Priyanka:~/ledger-api-assignment\$ kubectl get pods -n payments

NAME	READY	STATUS	RESTARTS	AGE
ledger-api-8577656ccc-6t2dh	2/2	Running	0	14m
ledger-api-8577656ccc-dvtzn	2/2	Running	0	14m
ledger-api-8577656ccc-t62bj	2/2	Running	0	13m
reporting-bc56f5ddb-jbr45	2/2	Running	0	9m11s

priya59@Priyanka:~/ledger-api-assignment\$ kubectl exec -it <REPORTING_POD_NAME> -n payments -- \

wget -qO- http://ledger-api:8080/health

-bash: REPORTING_POD_NAME: No such file or directory

priya59@Priyanka:~/ledger-api-assignment\$ kubectl exec -it reporting-bc56f5ddb-jbr45 -n payments -- \

wget -qO- http://ledger-api:8080/health

{"status": "ok"}

priya59@Priyanka:~/ledger-api-assignment\$ mkdir -p task-3/evidence

kubectl get pods -n payments > task-3/evidence/pods.txt

kubectl get peerauthentication -n payments -o yaml \
> task-3/evidence/peer-authentication.yaml

kubectl get authorizationpolicy -n payments -o yaml \
> task-3/evidence/authorization-policies.yaml

kubectl get networkpolicy -n payments \
> task-3/evidence/networkpolicy.txt

istioctl analyze -n payments \
> task-3/evidence/istio-analyze.txt

✓No validation issues found when analyzing namespace: payments.

priya59@Priyanka:~/ledger-api-assignment\$ |